

Managerial Decision Making
Business 38802 (AXP 25)
Autumn 2025
George Wu

Class 2

Judgment under Uncertainty:
Heuristics and Biases

Course Outline

Monday, November 3	Introduction to Decision Making
Tuesday, November 4	Judgment under Uncertainty I (Heuristics and biases)
Wednesday, November 5	Judgment under Uncertainty II (The difficulty of learning)
Friday, November 7	Risky Decision Making
Saturday, November 8	Nudges/Improving Decision Processes
January 5	Final (More later)

Today's Objectives

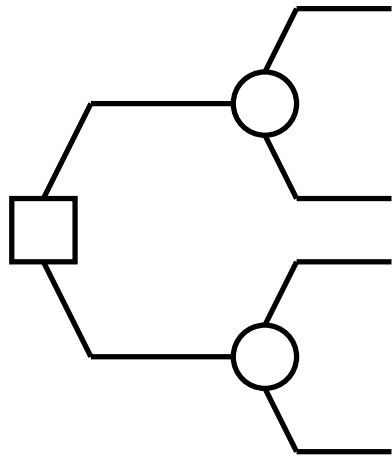
- Goals: Heuristics and Biases

- Important role of judgments in decision making
- Two effects: Overconfidence and optimism
- Two causes (for now): Anchoring and availability

- Outline

- Introduction
- Caselet: America Online
- Calibration exercise
- Lecture: Anchoring and availability
- Lecture: Updating judgments and Bayes Rule
- Prescriptions

Decision Trees



■ Alternatives

- Decision tree only includes identified alternatives (Nutt [1993] study of 168 corporate decisions; 28% had one option)

■ Beliefs

- What if the probabilities are systematically biased/noisy?
- What if you fail to identify a critical uncertainty?

■ Consequences

- How do/should you measure/weigh the outcomes?
- What if you fail to consider an important objective?

John Brown



“The core predicament of medicine – the thing that makes being a patient so wrenching, being a doctor so difficult, and being a part of a society that pays the bills they run up so vexing – is uncertainty ... Medicine’s ground state is uncertainty. And wisdom – for both patients and doctors – is defined by how one copes with it.”

- Atul Gawande (2002), *Complications* (“The case of the red leg”)

“Coping” with Uncertainty

■ Step 1

- Recognize that there is uncertainty
- Identify *critical* uncertainties



Some recent BIG decisions: how did you determine which uncertainties to consider? Did you miss any big ones? Why?

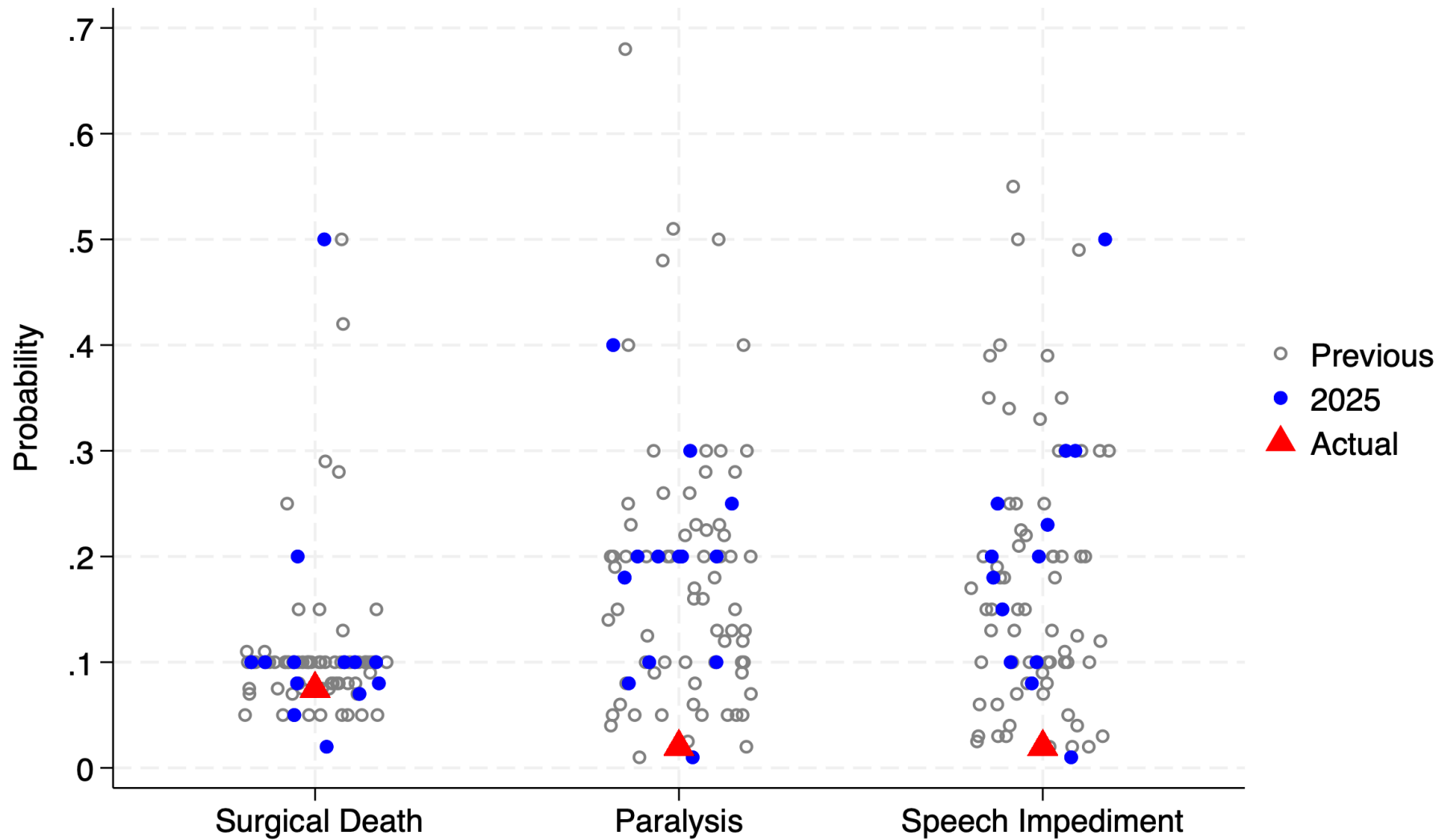
■ Step 2

- “Properly” reflect uncertainty in your judgments
- Collect information to inform you “appropriately”

■ Step 3

- Manage Uncertainty

John Brown Again



John Brown: Psychological Biases

- Availability

- Overweight outcomes of patients around you
 - Do you want a life/death decision to depend on whether you are sharing a room with someone who was lucky or unlucky?

- The “ $1/n$ heuristic”

- Informed consent: “Standard of care requires physicians to ‘inform a patient of the dangers of, possible negative consequences of, and alternatives to a proposed treatment or procedure.’ ” (Ciorte, 2017)
- Perhaps we start by anchoring on $1/n$
 - Low probability risks are over-estimated

- Note: Maybe doctors also are susceptible to these and other biases

“Coping” with Uncertainty: John Brown

■ Step 1

- Recognize that there is uncertainty
- Identify *critical* uncertainties
- John Brown: Only a few groups recognized that making sense of health outcomes (paralysis, speech impediment) is a forecast

■ Step 2

- “Properly” reflect uncertainty in your judgments
- Collect information to inform you “appropriately”
- John Brown: You did a crummy job
- John Brown: You need to refine “just enough”

■ Step 3

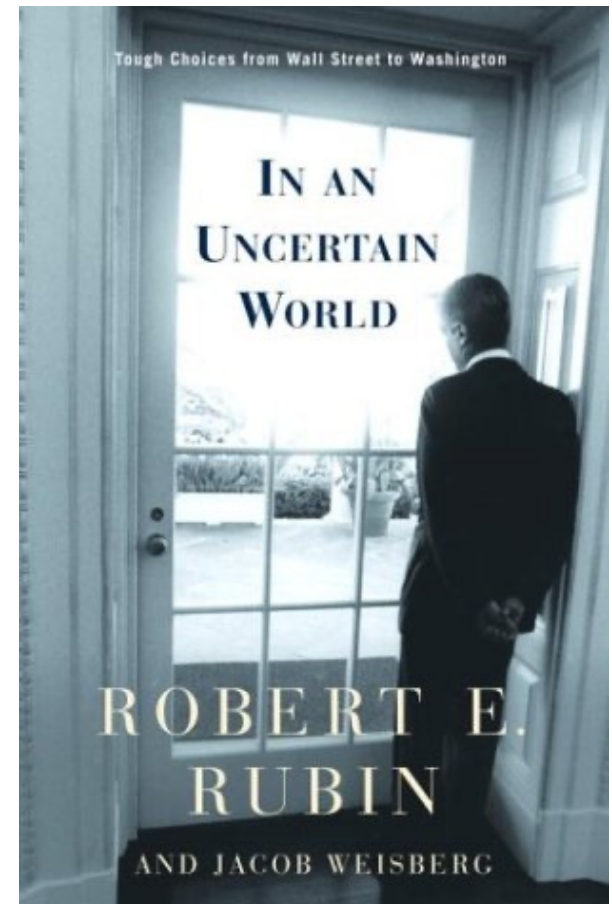
- Manage Uncertainty
-



Robert Rubin

...while a great many people accept the concept of probabilistic decision making and even think of themselves as practitioners, very few have internalized the mind-set. For me, probabilistic thinking has long been a highly conscious process. I imagine the mind as a virtual legal pad, with the factors involved in a decision gathered, weighed, and totaled up. To describe probabilistic thinking this way does not, however, mean that it can be reduced to a mathematical formula, with the best decision jumping automatically off a legal pad. Sound decisions are based on identifying relevant variables and attaching probabilities to each of them. That's an analytic process but also involves subjective judgments. The ultimate decision then reflects all of this input, but also instinct, experience, and "feel."

In an Uncertain World (2003)





Charlie Munger

If you don't get this elementary, but mildly unnatural mathematics of elementary probability into your repertoire, then you go through a long life like a one-legged man in an ass-kicking contest.

One of the advantages of a fellow like Buffet, whom I've worked with all these years, is that he *automatically* thinks in terms of decision trees and the elementary math of permutations and combinations...



 **AOL**.COM

America Online I

- October 29, 1996
 - AOL hires Bob Pittman and announces a pricing change
- Pricing Change
 - The Past: AOL has been charging:
 - \$19.95/month for 20 hours of service (plus \$2.95/additional hour) or
 - \$9.95/month for five hours (plus \$2.95/additional hour)
 - Effective December 1, 1996: AOL will introduce a new flat-rate structure
 - \$19.95/month for unlimited access

America Online II

- What are the critical uncertainties?
- How would you resolve these uncertainties?

America Online III

- So, what would you do?

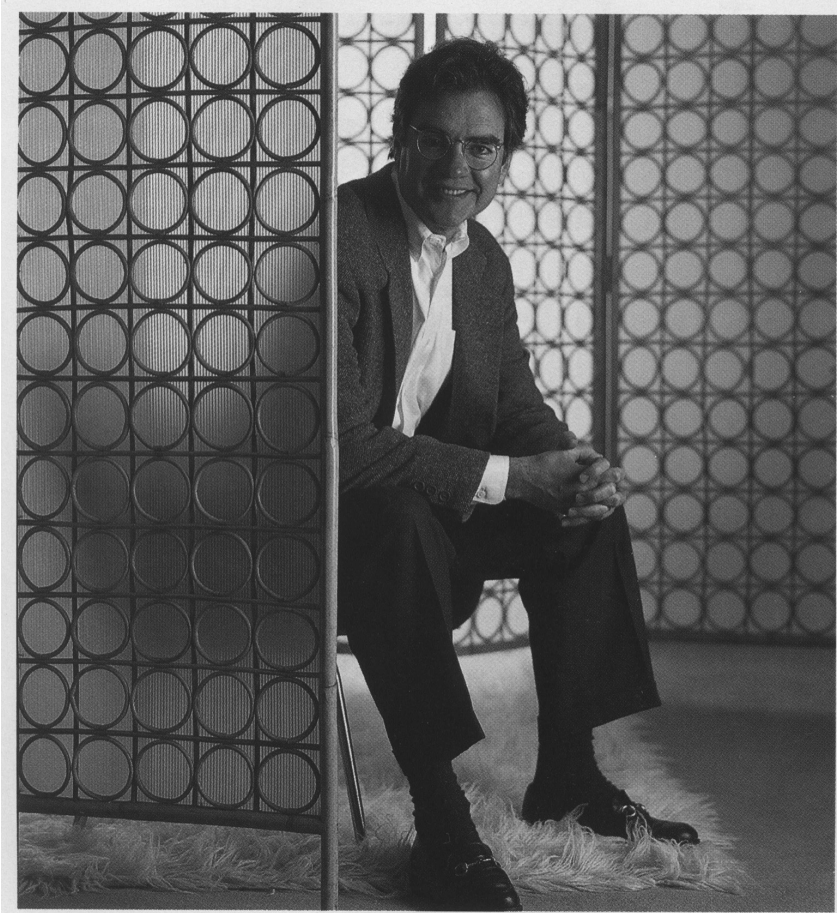
America Online: Planning

- ❑ Began bracing in September
 - November: added 12,000 new modems (to 260,000 existing modems) (along with more phone lines and computer servers).
- ❑ Steve Case: AOL would “really be challenged to meet that demand and try to avoid busy signals and system sluggishness.”
- ❑ David Gang, VP marketing: “We’re going into territory that nobody on the face of the planet has ever been before.”

America Online: What Happened

- ❑ Peak hour logins skyrocketed from 140,000 in early October to 236,000 the week before Christmas.
- ❑ Average time spent online by AOL subscriber doubled since September, rising from 14 minutes to 32 minutes.
- ❑ CompuServe Super Bowl ad: 15 seconds of blank screen and repeated (and unsuccessful) attempts to login
https://www.youtube.com/watch?v=-xH0CsZ_TnE
 - 1-888-NOTBUSY
- ❑ Case urged AOL members who also subscribe to other Internet services to use them as back doors into the AOL system.

America Online: Epilogue



Bob Pittman, President of AOL:

Forecasts are fairly reliable if you have a historical precedent in any business or opportunity. They get tricky if you have no historical precedent. No one has ever been our size before in this business. There is nothing else in the world like AOL and there never has been. Who would have thought usage would double. I mean, that's like saying the ratings of a radio station or a TV station doubled in one month.

Primary versus Secondary Knowledge

- Primary knowledge

- Trivia Pursuit test: “How much do you know?”

- Secondary knowledge

- Confucius test: “How much do you think you know?”

- “Shall I tell you what it is to know? To say you know what you know, and to say you do not know when you do not, that is knowledge.”

- Why study secondary knowledge?

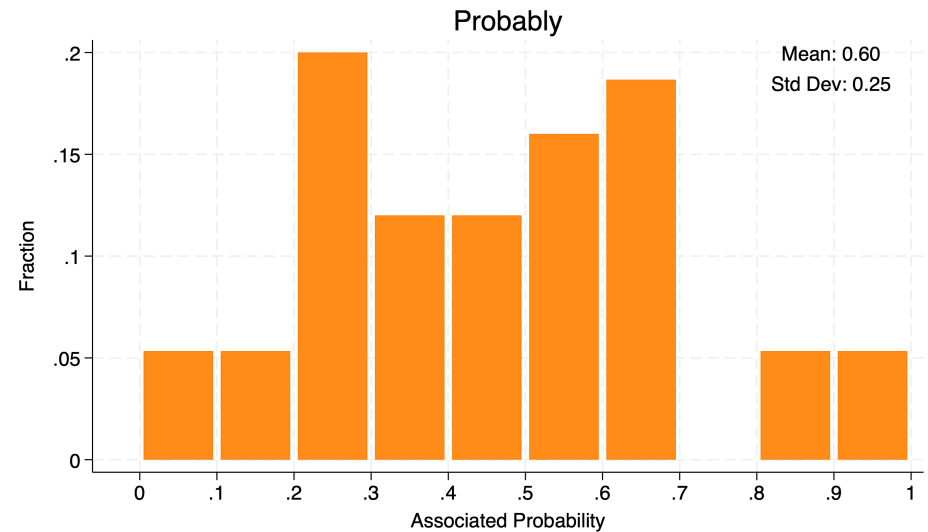
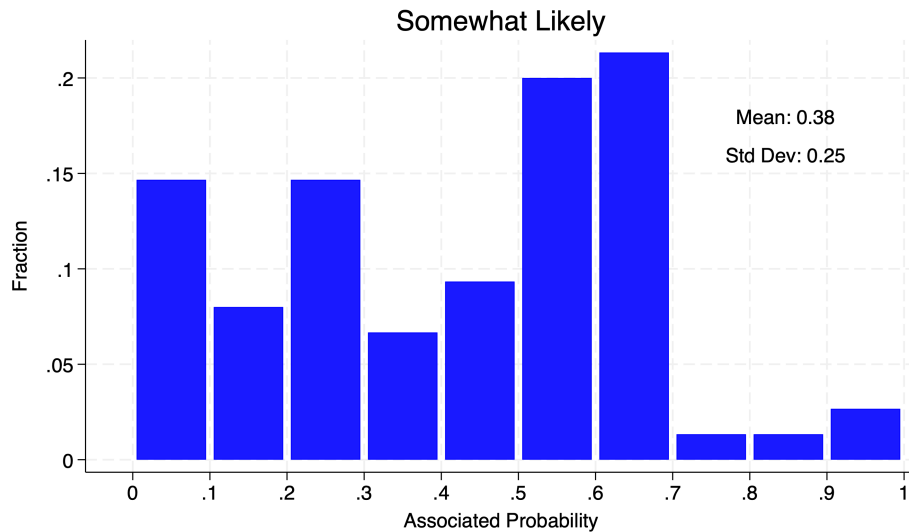
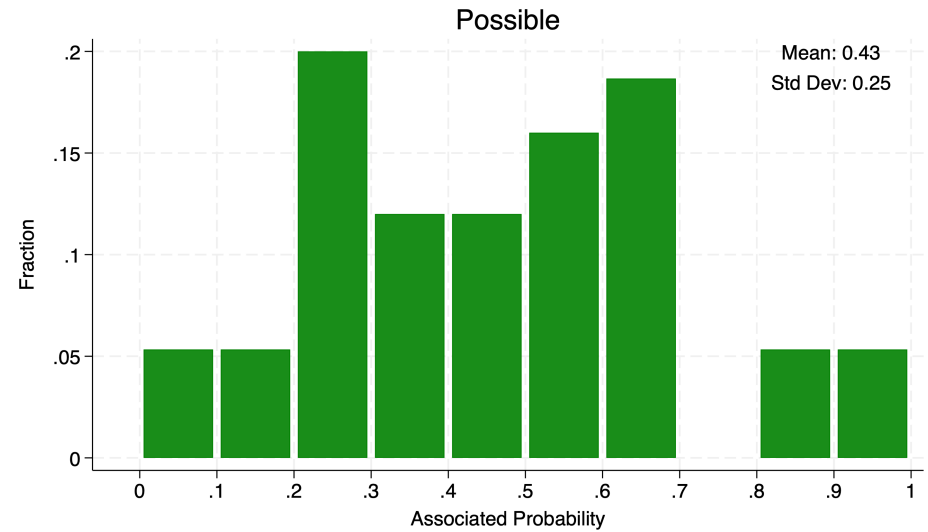
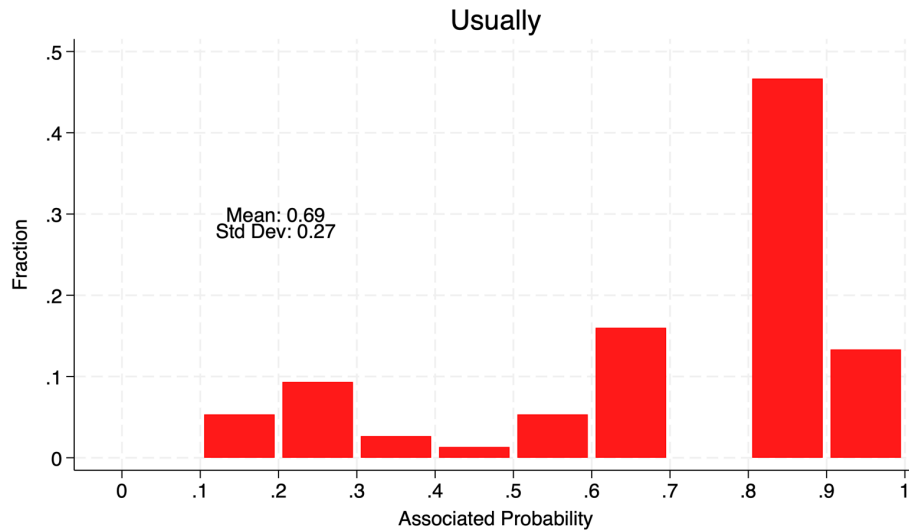
- Make better decisions
 - Underestimate risk
 - Fail to collect information
 - Miss risk management opportunities

Quantifying our uncertainty

What probability would you assign to the following verbal statements?

1. “Usually”
2. “Possible”
3. “Somewhat likely”
4. “Probably”

Quantifying Uncertainty: Your Data



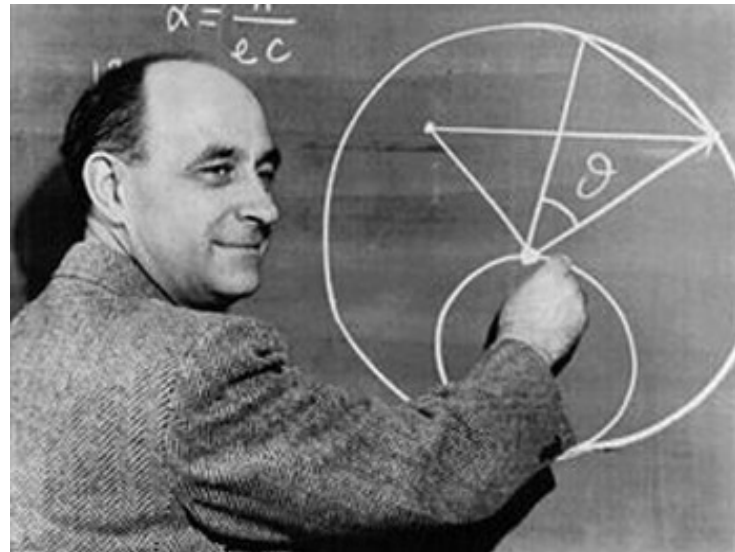
Why quantify our uncertainty?

Rank	Statement	N	Mean	Median	σ	Range	Rank	Statement	N	Mean	Median	σ	Range
1	Highly probable	187	0.89	0.90	0.04	.60-.99	21	Slightly less than	188	0.45	0.45	0.04	.05-.50
2	Very likely	185	0.87	0.90	0.06	.45-.99		half the time					
3	Very probably	187	0.87	0.89	0.07	.60-.99	22	Slight odds against	185	0.45	0.45	0.11	.10-.99
4	Quite likely	188	0.79	0.80	0.10	.30-.99	23	Not quite even	180	0.44	0.45	0.07	.05-.60
5	Usually	187	0.77	0.75	0.13	.15-.99	24	Inconclusive	153	0.43	0.50	0.14	.01-.75
6	Good Chance	188	0.74	0.75	0.12	.25-.96	25	Uncertain	173	0.40	0.50	0.14	.01-.90
7	Predictable	146	0.74	0.75	0.20	.01-.99	26	Possible	178	0.37	0.49	0.23	.01-.99
8	Likely	188	0.72	0.75	0.11	.25-.99	27	Somewhat unlikely	186	0.31	0.33	0.12	.03-.80
9	Probably	188	0.71	0.75	0.17	.01-.99	28	Fairly unlikely	187	0.25	0.25	0.11	.02-.75
10	Rather likely	188	0.69	0.70	0.09	.15-.99	29	Rather unlikely	187	0.24	0.25	0.12	.01-.75
11	Pretty good chance	188	0.67	0.70	0.12	.25-.95	30	Not very probably	187	0.21	0.20	0.12	.01-.60
12	Fairly likely	188	0.66	0.70	0.12	.15-.95	31	Unlikely	188	0.18	0.16	0.10	.01-.45
13	Somewhat likely	187	0.59	0.60	0.18	.20-.92	32	Not much chance	186	0.16	0.15	0.09	.01-.45
14	Better than even	187	0.58	0.60	0.06	.45-.89	33	Seldom	188	0.16	0.15	0.08	.01-.47
15	Rather	124	0.58	0.60	0.11	.10-.80	34	Barely possible	180	0.13	0.05	0.17	.01-.60
16	Slightly more than	188	0.55	0.55	0.06	.45-.80	35	Fairly possible	184	0.13	0.05	0.16	.01-.60
	half the time						36	Improbably	187	0.12	0.10	0.09	.01-.40
17	Slight odds in favor	187	0.55	0.55	0.08	.05-.75	37	Quite unlikely	187	0.11	0.10	0.08	.01-.50
18	Fair chance	188	0.51	0.50	0.13	.20-.85	38	Very unlikely	186	0.09	0.10	0.07	.01-.50
19	Tossup	188	0.50	0.50	0.00	.45-.52	39	Rare	187	0.07	0.05	0.07	.01-.30
20	Fighting chance	186	0.47	0.55	0.17	.05-.90	40	Highly improbable	181	0.06	0.05	0.05	.01-.30

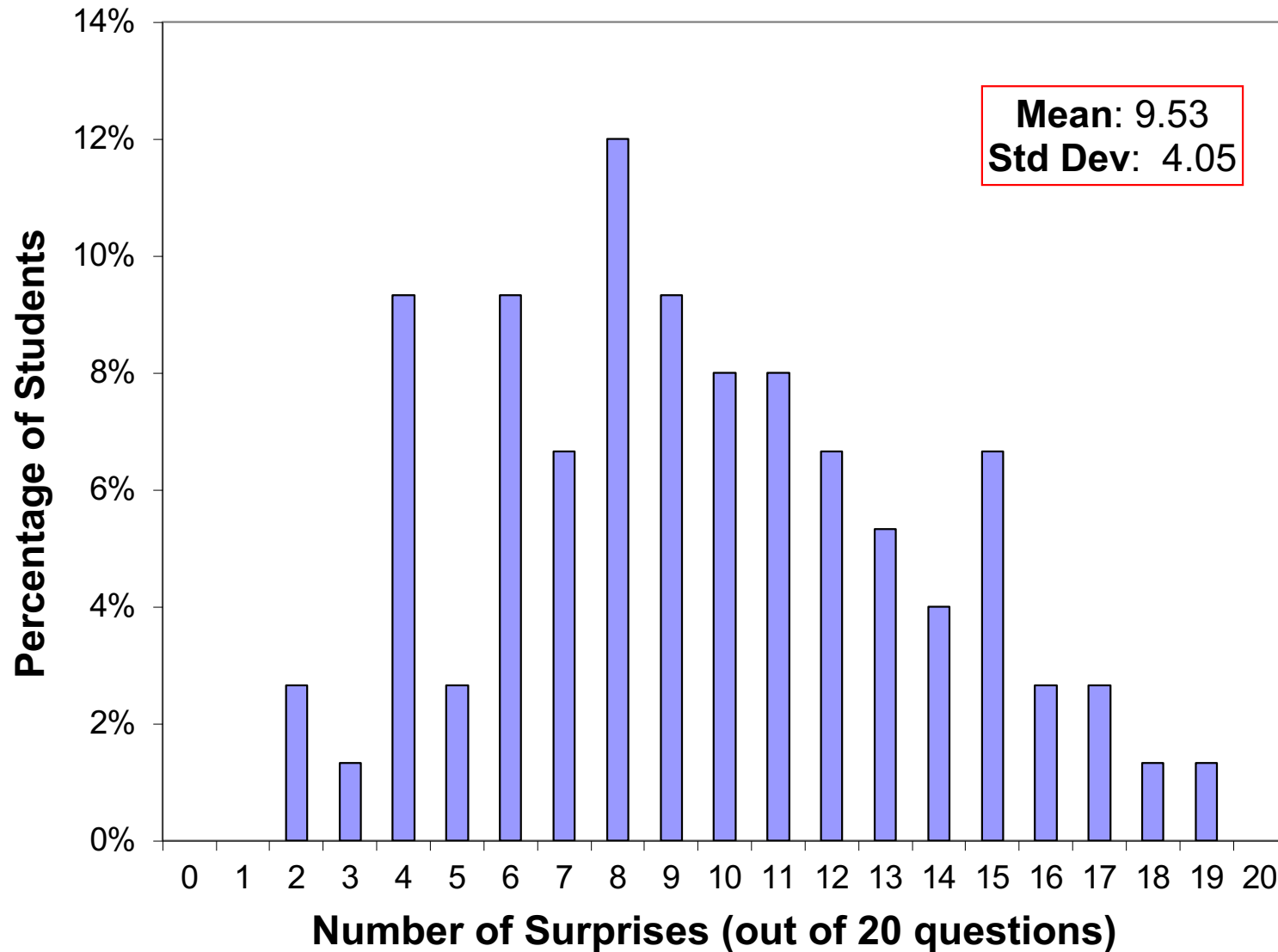
Probability Exercise



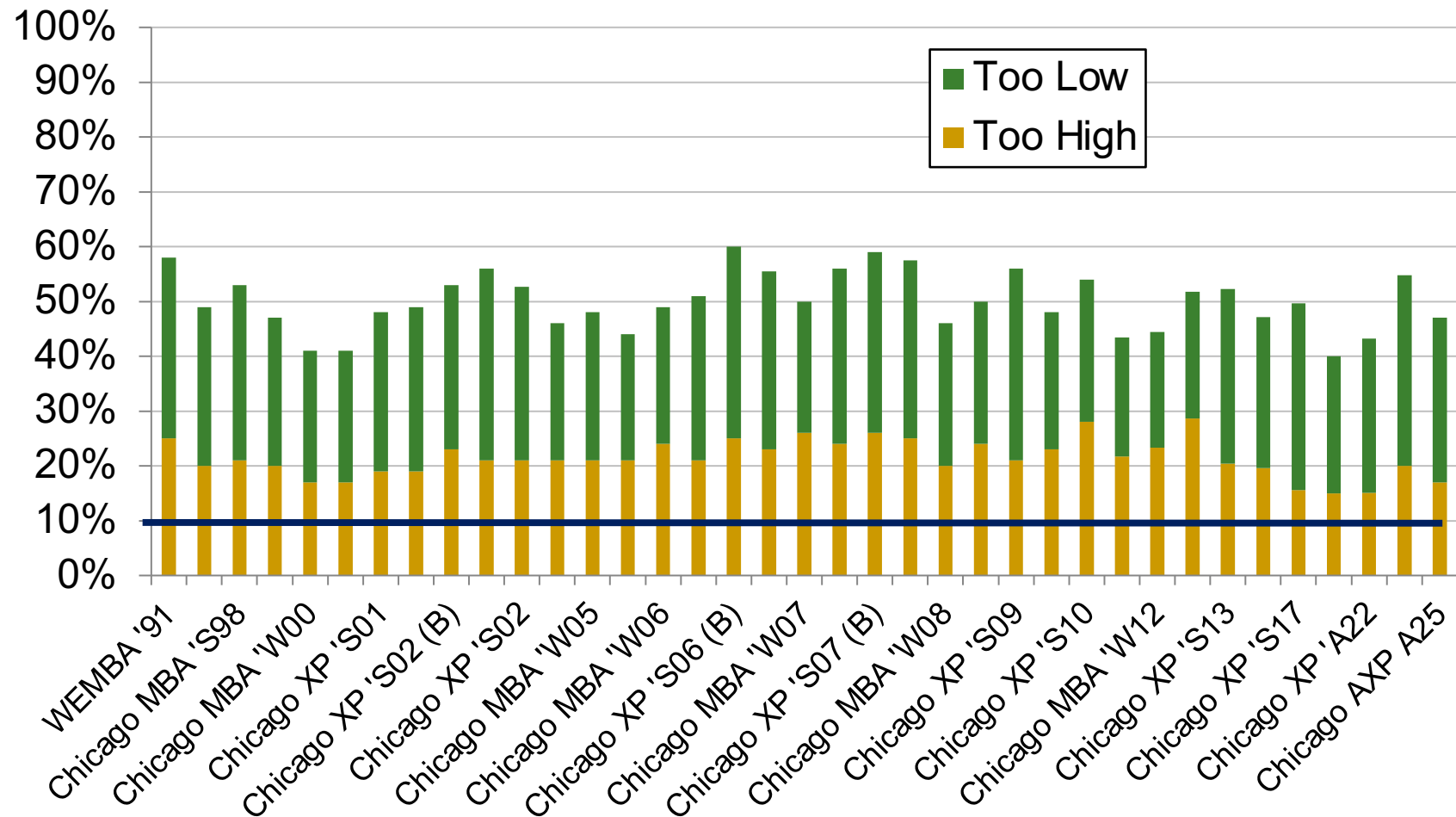
- “Bezos is famous for asking candidates brain teasers like ‘How many windows are in the city of San Francisco?’ or ‘How many trees are in New York’s Central Park?’ ” *Fortune* (May 12, 2003)



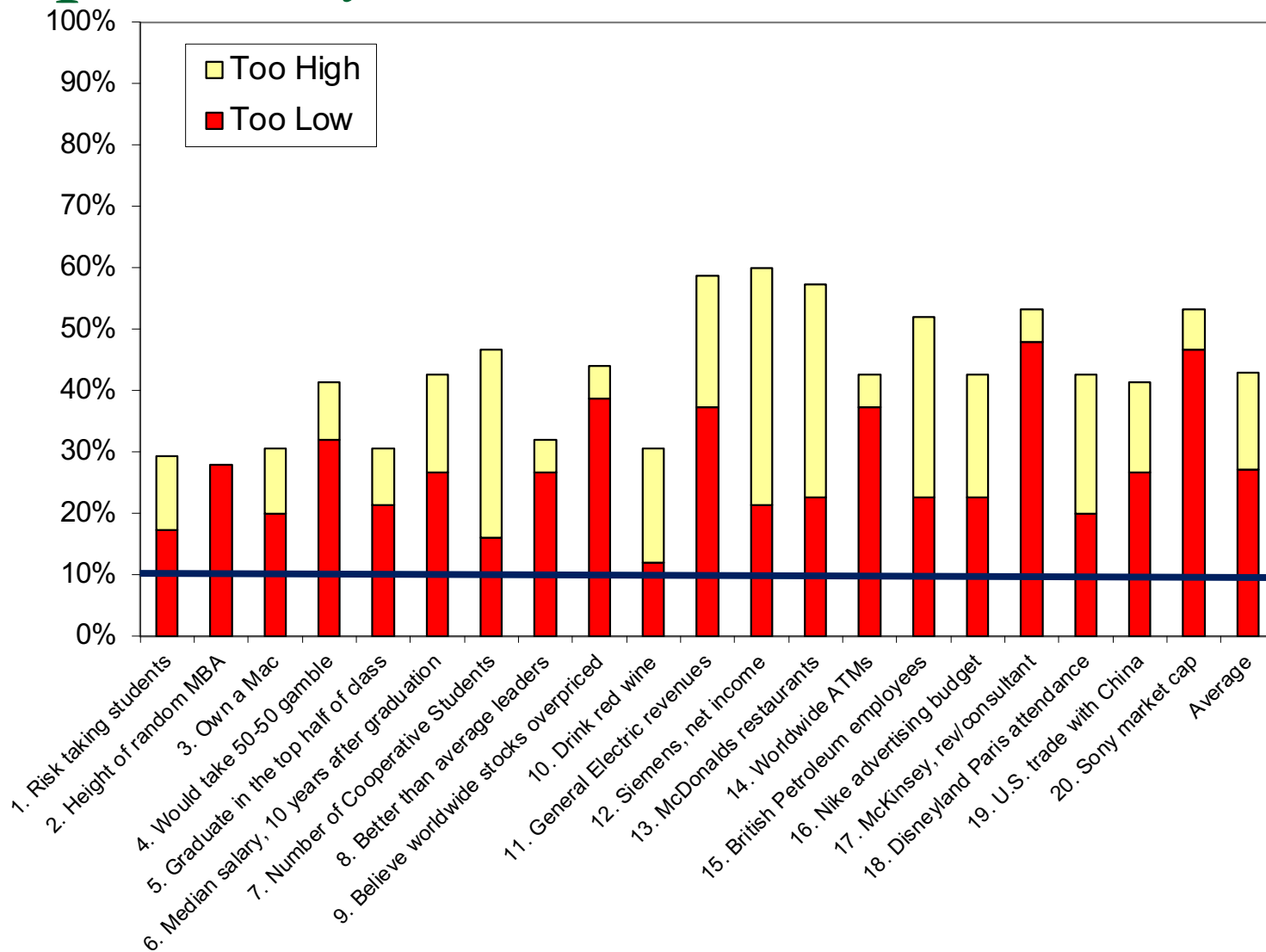
Surprises by Student



Calibration and Probability Assessment



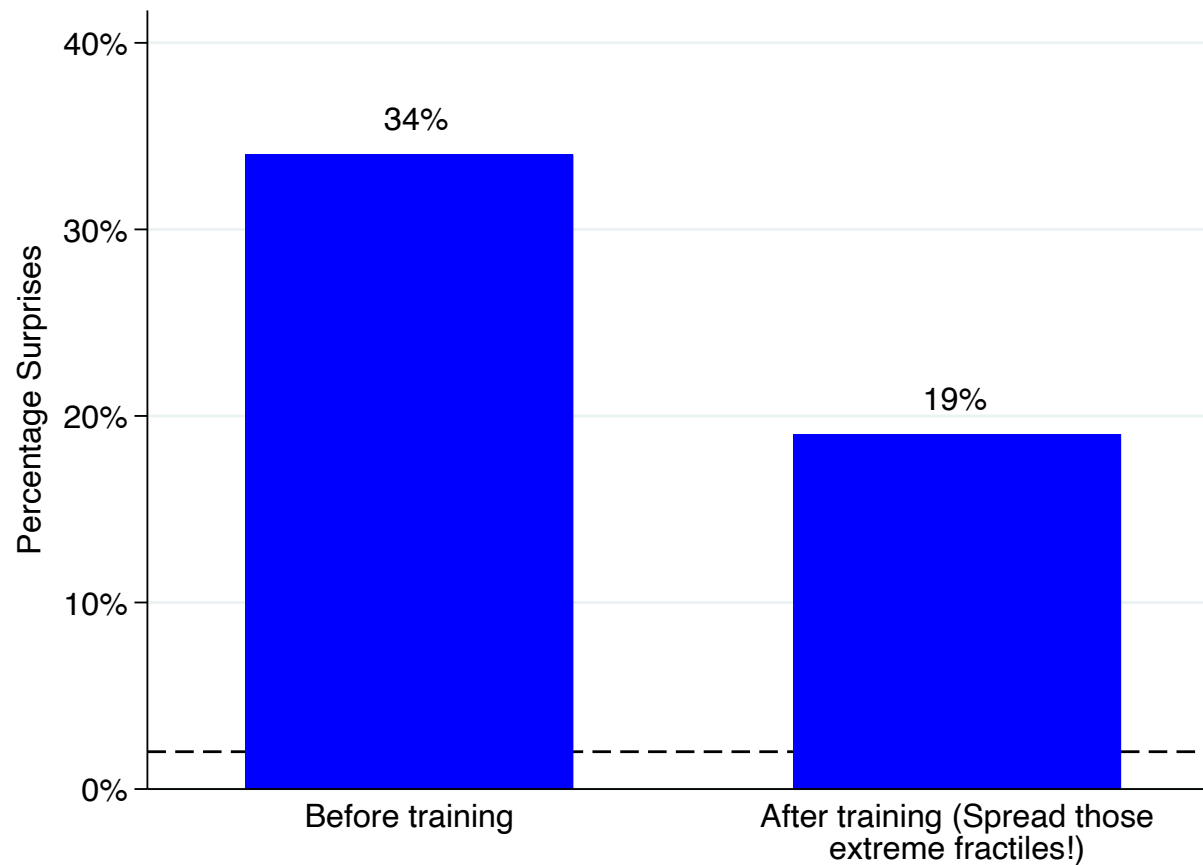
Surprises by Question



Calibration continued

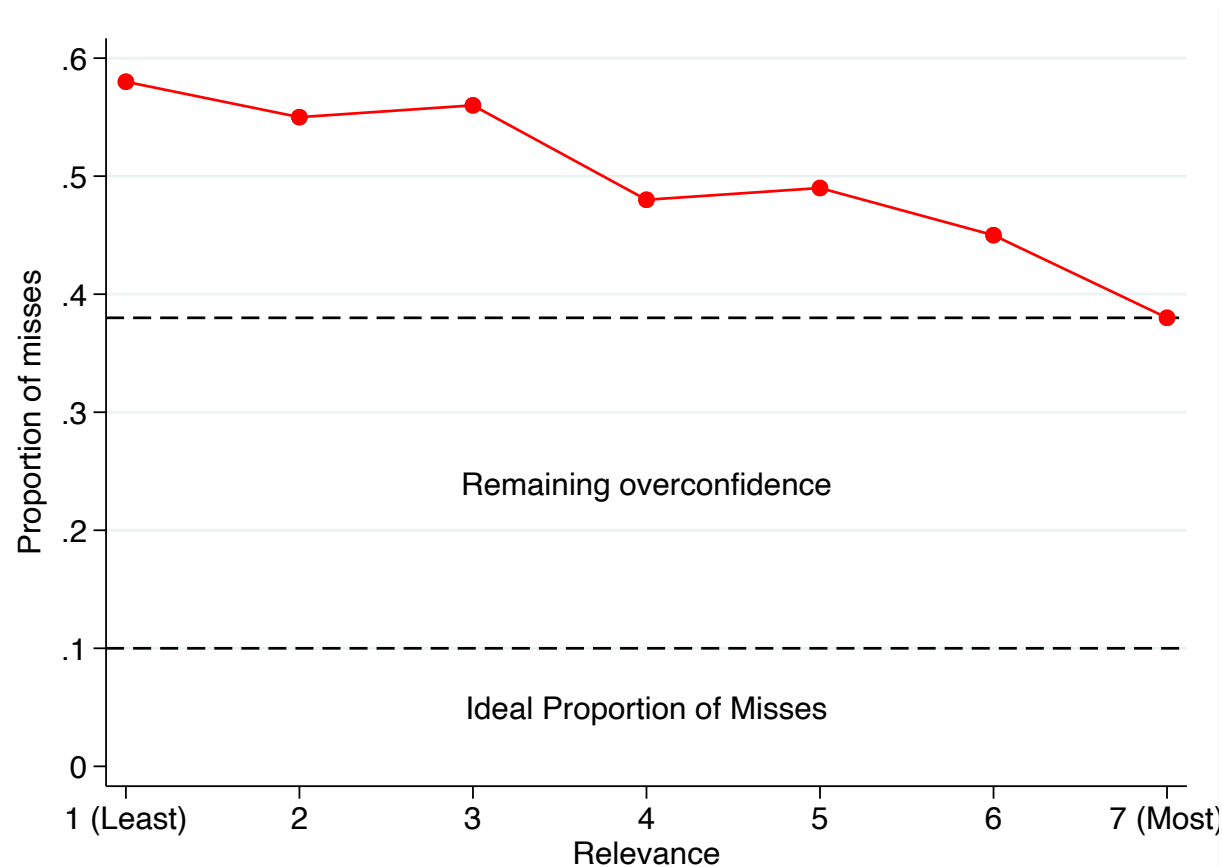
Instruction helps but does not eliminate problem

- Subjects: Harvard MBA students (Alpert & Raiffa)



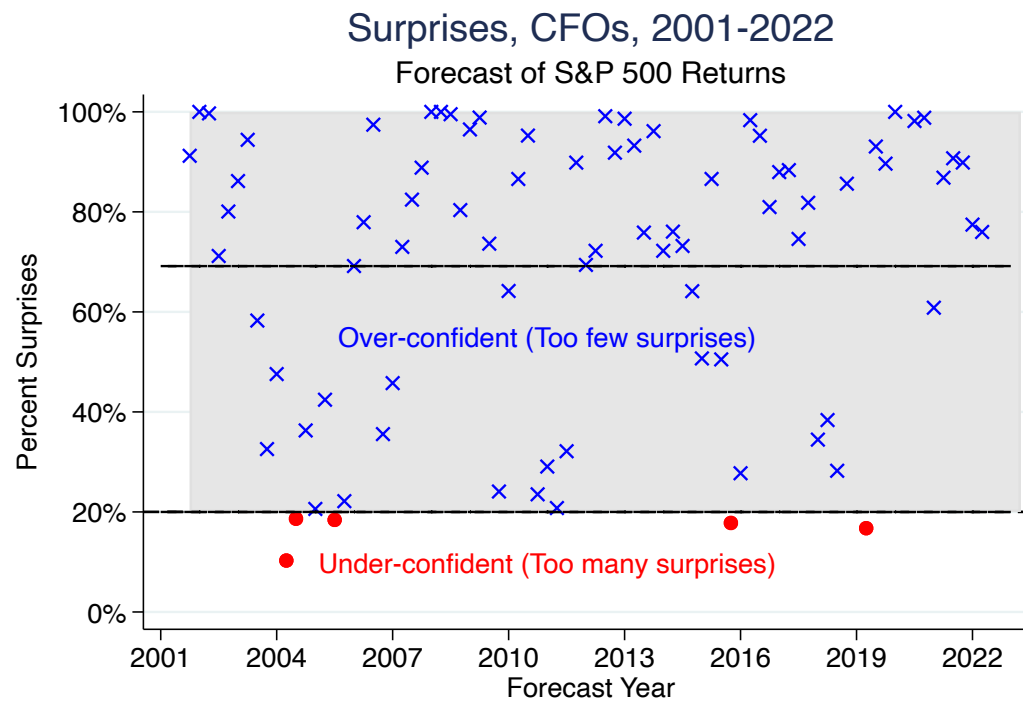
Calibration continued

Relevance of questions to job only partially reduces overconfidence (Russo & Schoemaker)



Overconfidence in CFOs

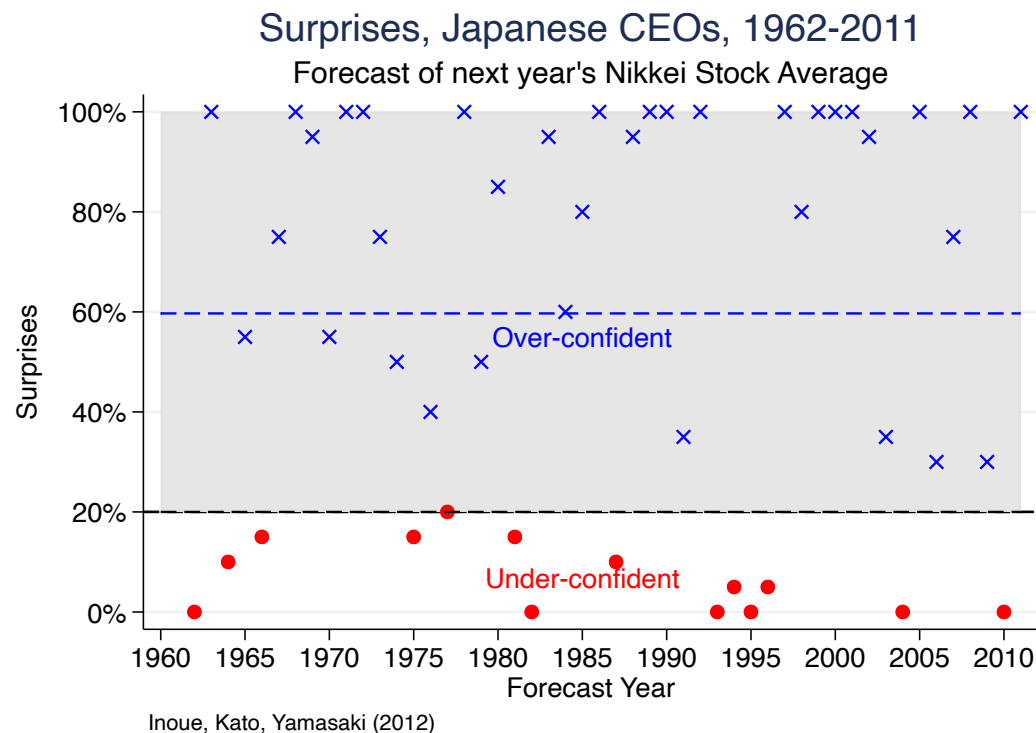
- Ben-David, Graham, & Harvey (2013)
 - Data: Survey of 2000-3000 CFOs, quarterly from 2001-2022
 - Responses: CFOs provide 1- and 10-year S&P 500 returns
 - Confidence Interval, using 10% and 90% fractiles



Boutrous, Ben-David, Graham, Harvey, and Payne (2024)

Overconfidence in CEOs

- Inoue, Kato, & Yamasaki (2012)
 - Data: Survey of 20 CEOs, annually from 1962-2011
 - Responses: CEOs provide 1-year forecast of Nikkei stock average
 - Confidence Interval, using “low” and “high” price



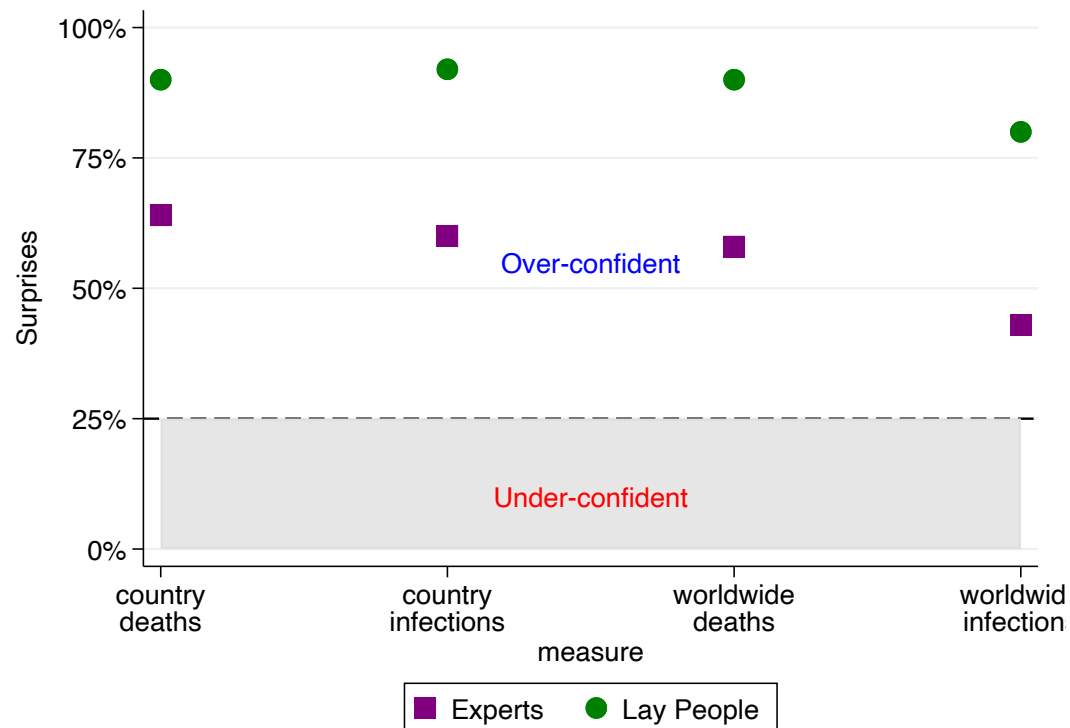
Overconfidence with COVID

- Recchia, Freeman, and Spiegelhalter (2021)

Data: Forecasts of COVID outcomes in the UK

- Responses: 140 Experts and 2,086 Lay people made 4 forecasts

- Confidence Interval, using 12.5% and 87.5% fractiles



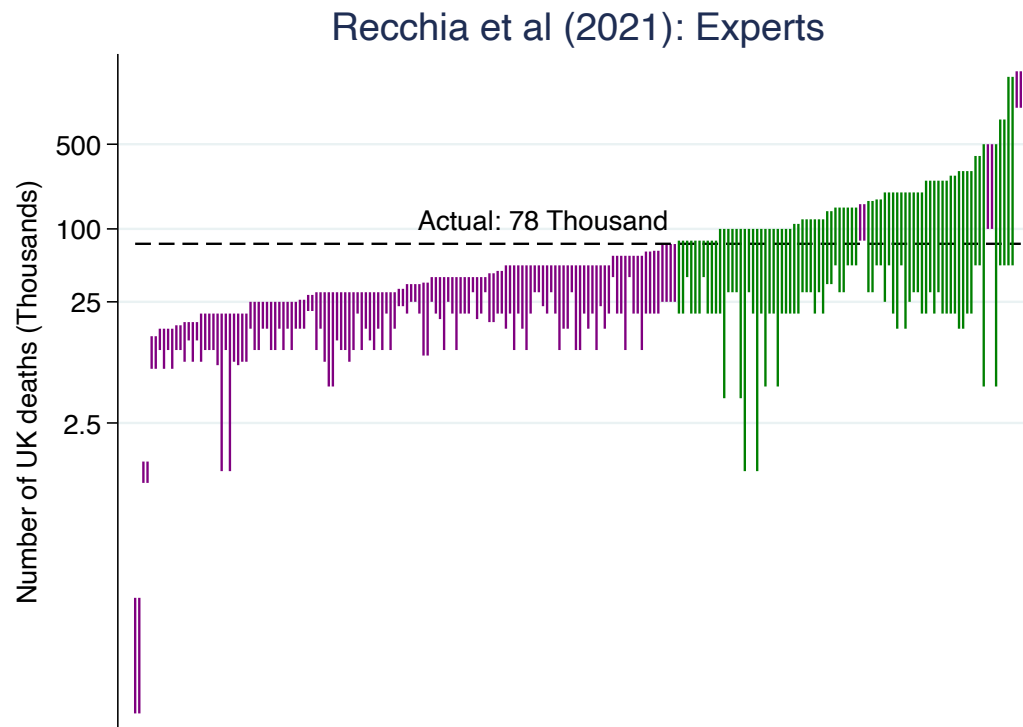
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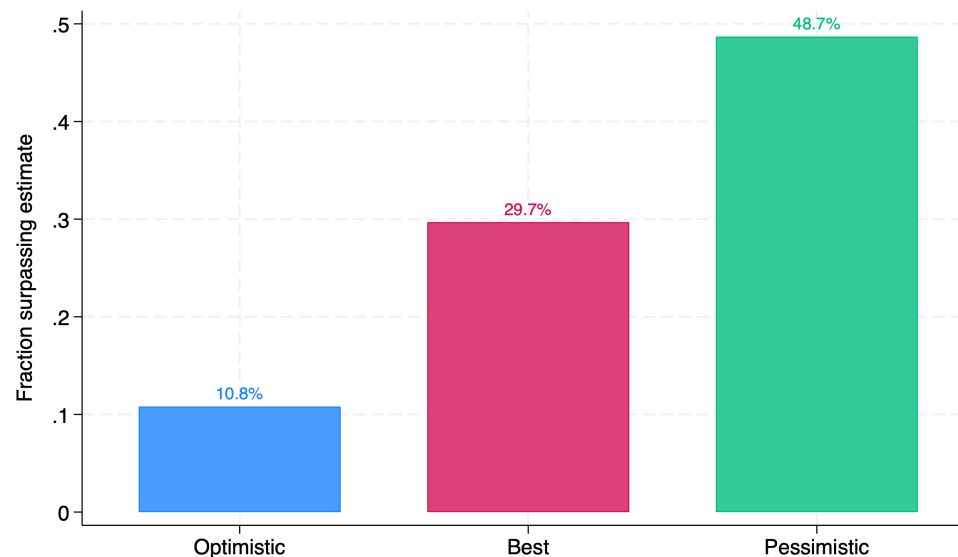


Overconfidence in Planning

- Buehler, Griffin, and Ross (1994)

Data: 37 psychology honor students made forecasts of timing of thesis completion

- Best estimate
- Optimistic (“if everything went as well as it possibly could”)
- Pessimistic (“if everything went as poorly as it possibly could”)



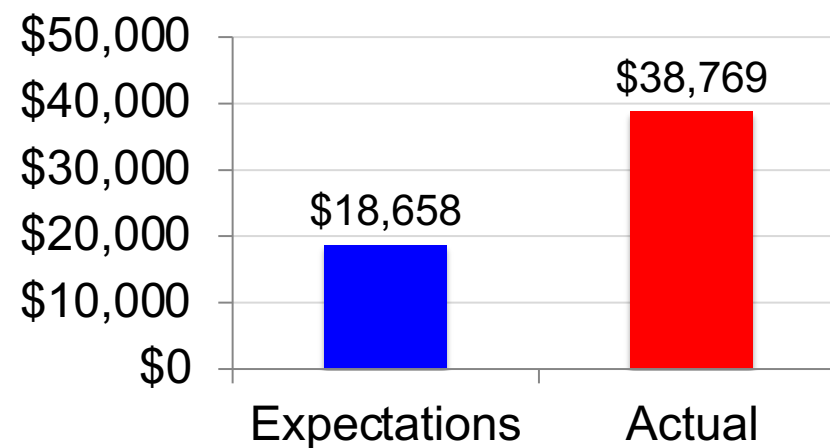
Optimism/Overconfidence Examples

■ Corporate takeovers

- high bids seem to be partly because acquiring firms
 - overestimate synergies
 - overestimate their capabilities as managers
 - overestimate the amount of control they will have

■ Project management

- almost all project constructions exceed initial cost and time estimates by over 20% (norm is twice)
- *Remodelling* (2002): Kitchen renovation



Forecasting is Hard!

- The Americans have need of the telephone, but we do not. We have plenty of messenger boys.
 - Sir William Preece, British Post Office, 1876



Forecasting is Hard!

- Everything that can be invented has been invented
 - Charles Duell, Commissioner, US Patent Office, 1899



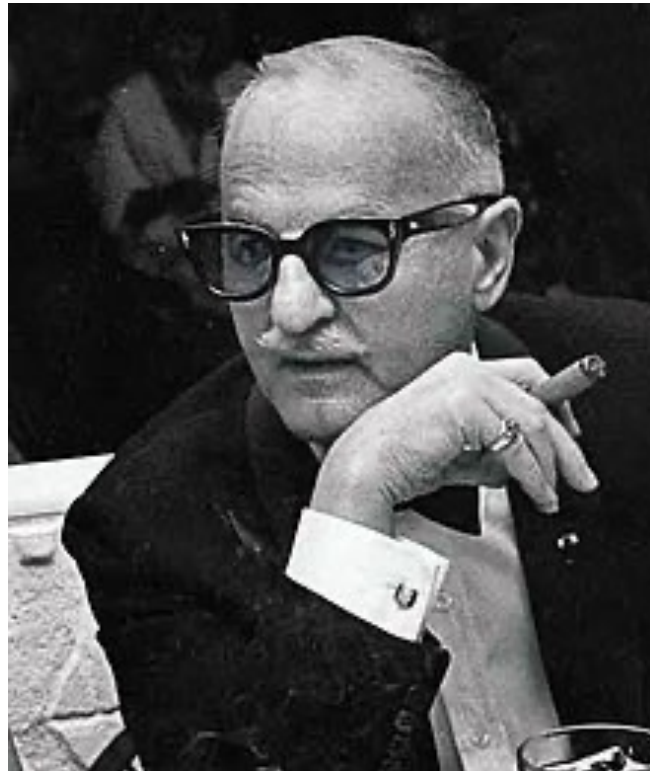
Forecasting is Hard!

- I think there's a world market for about five computers.
 - Thomas J. Watson, Chairman of IBM, 1943



Forecasting is Hard!

- TV won't be able to hold on to any market it captures after the first six months. People will soon get tired of staring at a plywood box every night.
 - Daryl F. Zanuck, Chairman 20th Century Fox, 1946



Forecasting is Hard!

- There's no chance that the iPhone is going to get any significant market share.
 - Steve Ballmer, Microsoft CEO, 2007



The “Planning Fallacy”



- Plans
 - ❑ Cost: AUS\$7 MM | Time: 4 years
- Actual
 - ❑ Cost: AUS\$105 MM | Time: 14 years

The “Planning Fallacy”

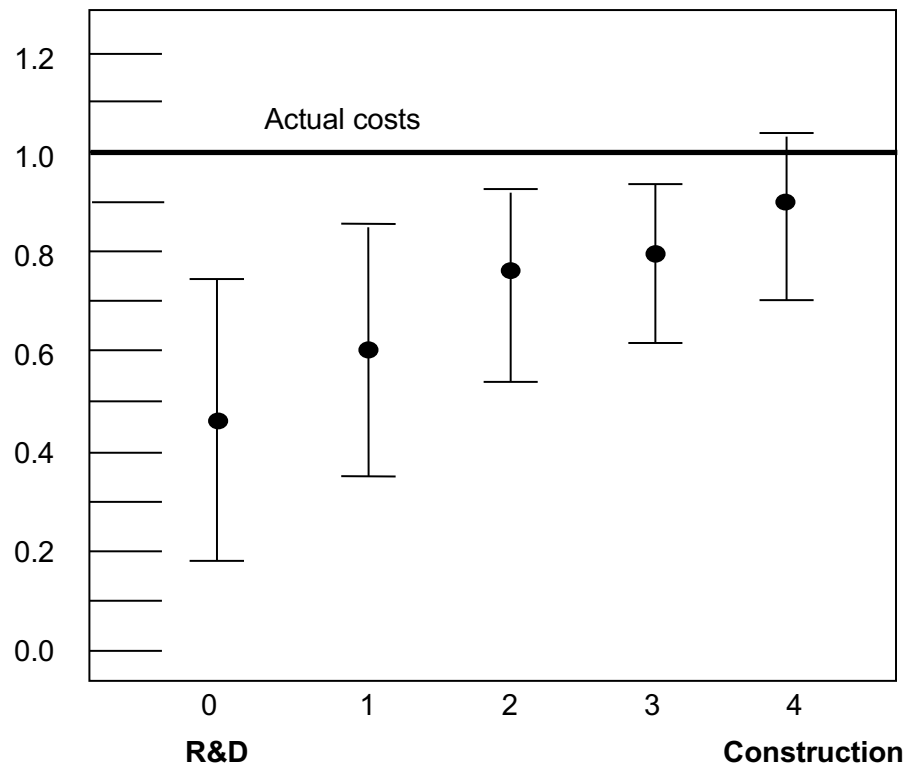
- Hong Kong XRL (26 kilometer high speed rail from HK to Guangzhou and Shenzhen)



- Forecasted Schedule: 2011-2015
- 2015: less than half work was completed
- 2018: completed (more later)

The “Planning Fallacy”

Study of Pioneer Process Plants (Merrow, Phillips and Meyers, 1981)



- At every stage of the process
- managers underestimate the cost of completing the construction
 - typical initial estimate is less than half the final cost.



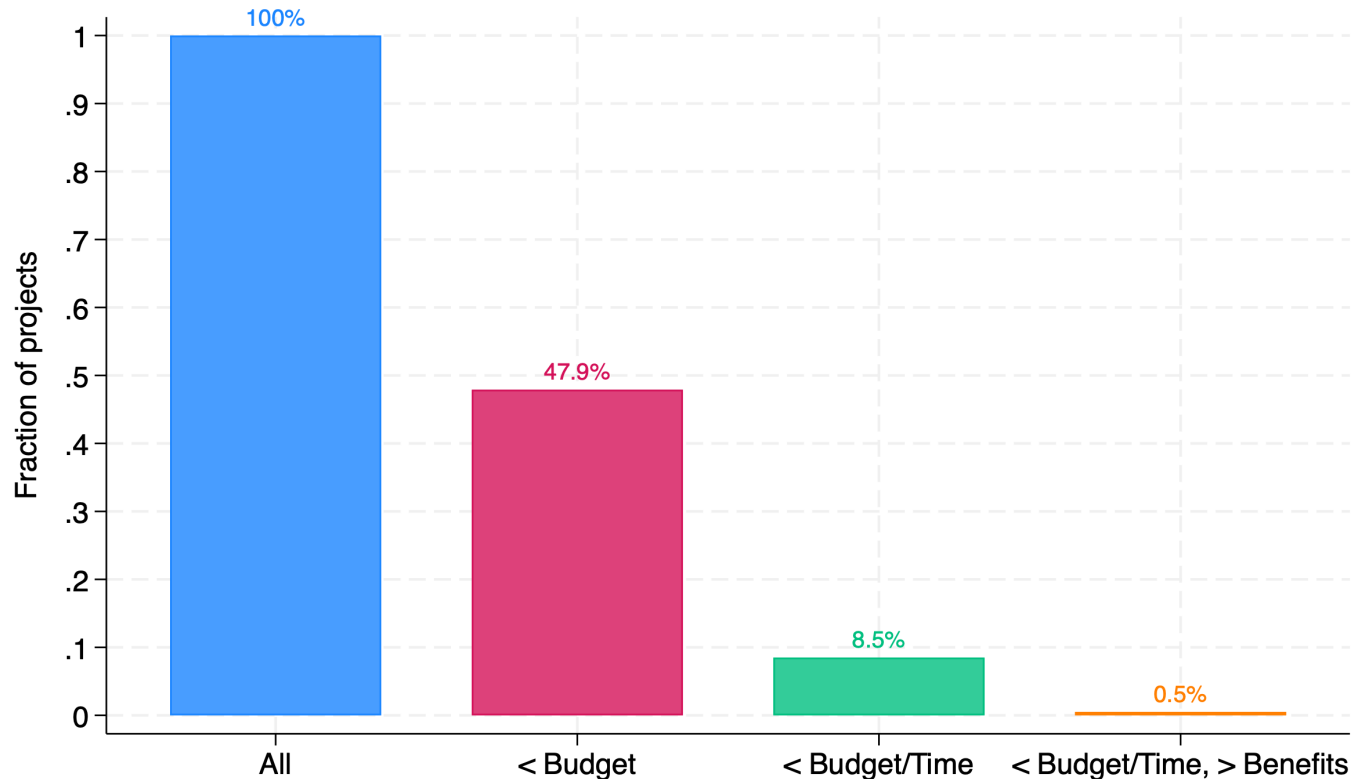
Planning Fallacy in your organization?

The “Planning Fallacy”

- Flyvberg and Gardner (2023)

Data: 16,000+ “megaprojects” (over \$1 billion)

- Budget, Time, Benefits



System 1 versus System 2

■ System 1

- ❑ Automatic, low effort processing (e.g. facial recognition)
- ❑ Characterized by: cognitive short-cuts, heuristics, first impressions, importance of context, etc.

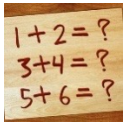
■ System 2

- ❑ Effortful, Purposeful (e.g., $\sqrt{19,305}$ or)
- ❑ Characterized by: careful cognitive processing, rule-based thinking, etc.

Both can operate in parallel. Often we have a System 1 response first, and then System 2 overrides. But, System 2 often uses System 1 judgments as inputs.

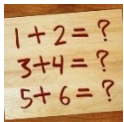
Anchoring and Adjustment

- We start with an initial value (a “best guess”) and then adjust, but...
 - Adjustments are insufficient (thus ranges are too narrow)...
 - ... thus, we are overconfident about what we know.



1+2=?
3+4=?
5+6=?

Spell out how anchoring leads to overly narrow confidence intervals?



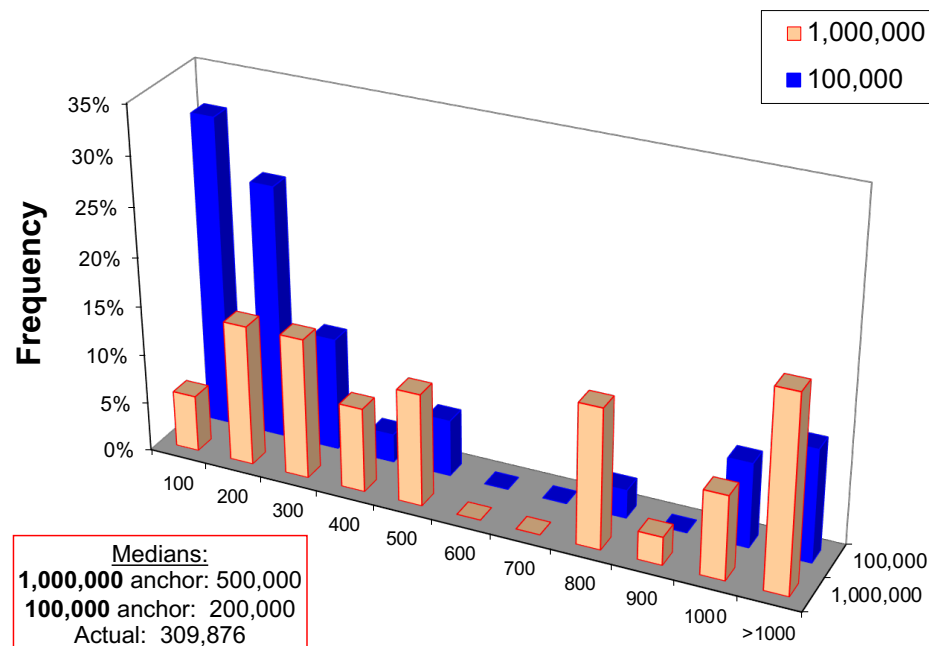
1+2=?
3+4=?
5+6=?

What role does anchoring play in the “1/n” heuristic?

Anchoring and Adjustment I

■ Example 1

- What is the probability that 2024 Toyota Camry passenger car sales (in United States) were higher than **100,000** (**1,000,000**)?
- What is your best guess (in thousands of cars) as to 2024 Toyota Camry passenger car sales (in United States)?

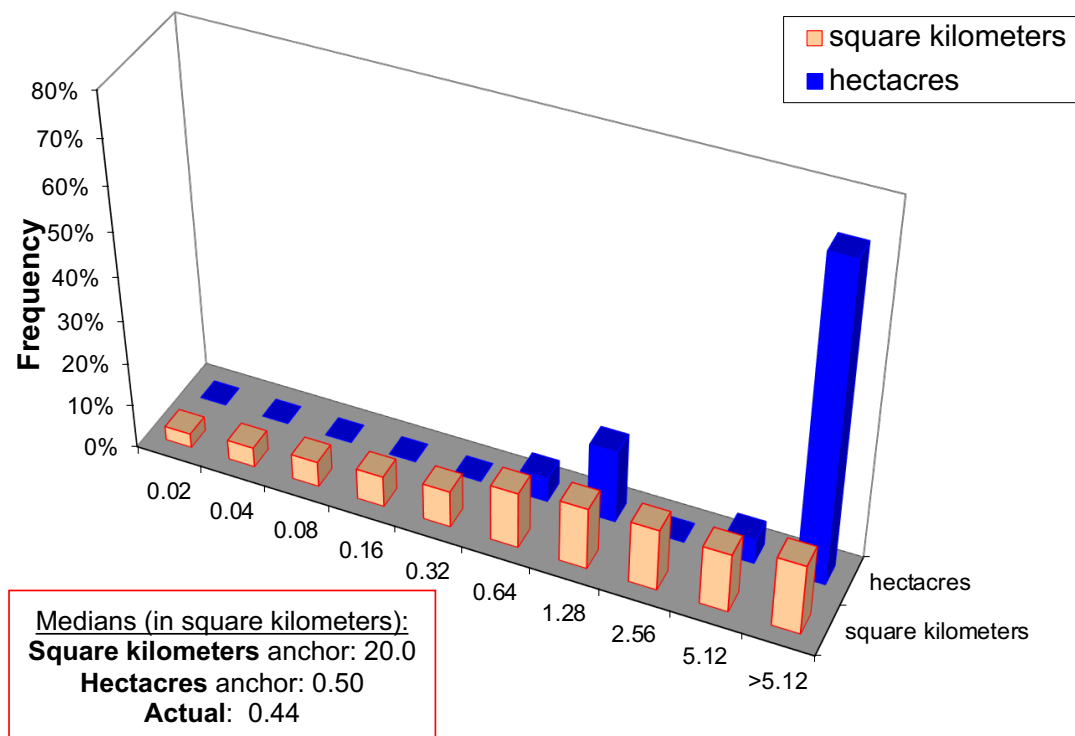


Toyota Camry 2024 Sales (thousands)

Anchoring and Adjustment II

■ Example 2

- Estimate the area of the Vatican City
 - in **square kilometers**
 - in **hectares**

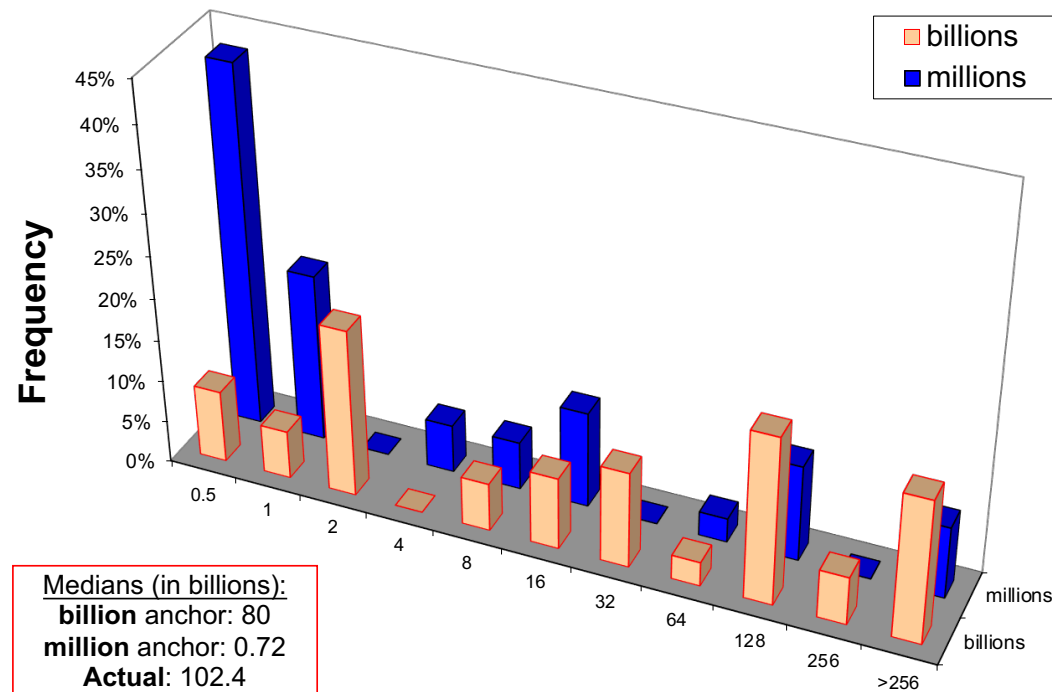


Interior area of Vatican (in square kilometers)

Anchoring and Adjustment III

■ Example 3

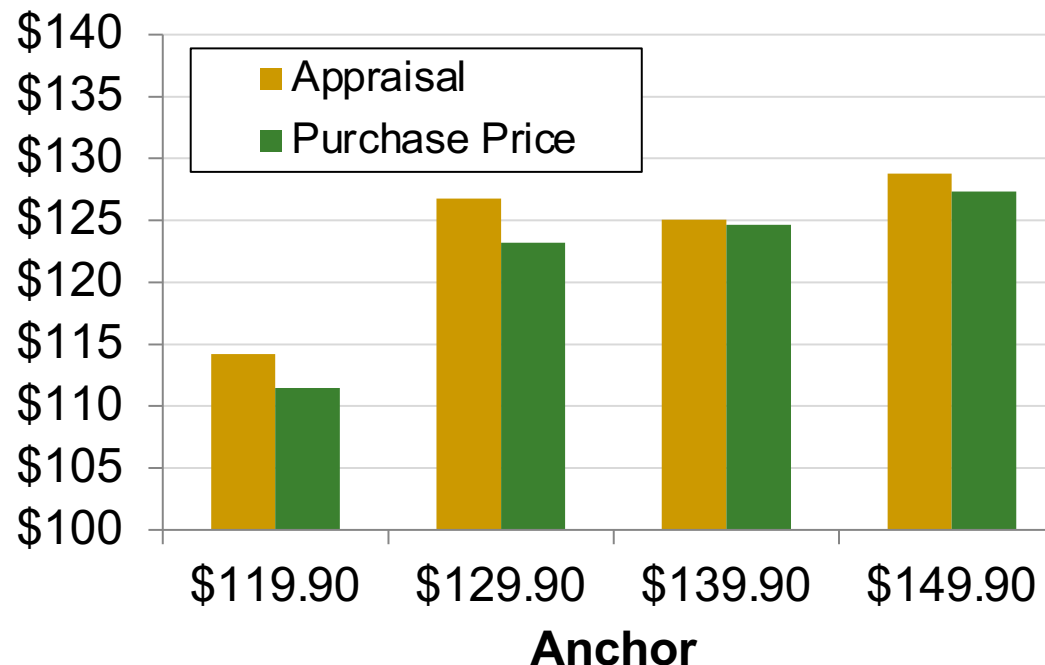
- Estimate the total egg production in 2024 in India
 - in **billions**
 - in **millions**



2024 Indian egg production (billions)

Anchoring and Adjustment IV

- Study of realtors (Northcraft & Neale, 1987)
 - Subjects: 47 Tucson realtors (mean experience: 8.9 years)
 - Judgments: Appraisal value and purchase price
 - Single property but 4 different list price anchors
 - Realtors: “List price is irrelevant”



Availability

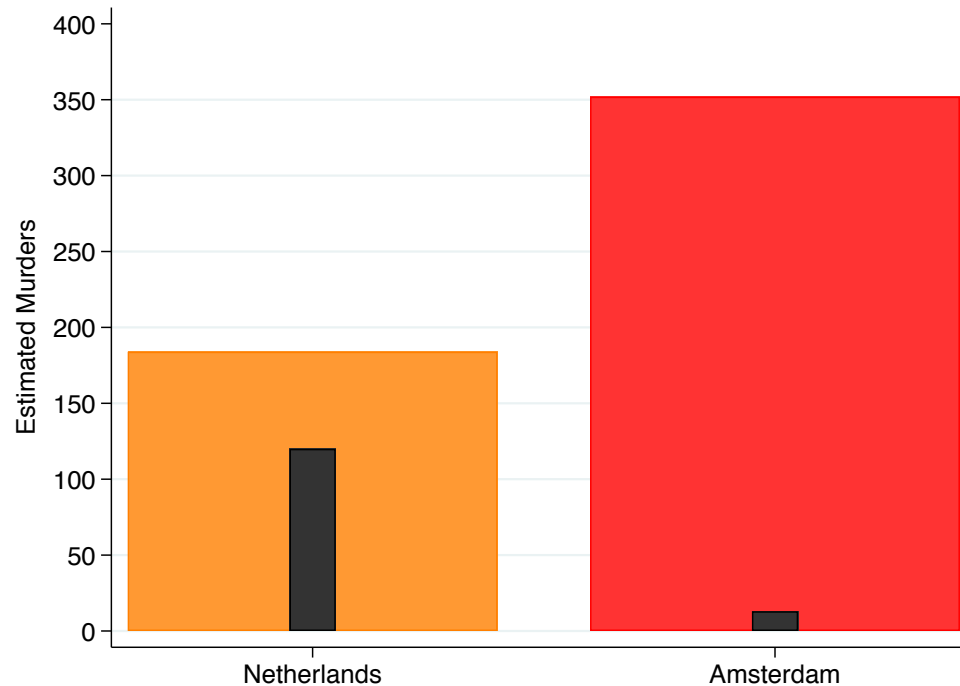
- In forming a judgment, we tend to
 - ❑ rely on readily available (recent, salient, vivid) information
 - ❑ but fail to discount the quality of the information, or consider other possible sources

Availability 0

■ UK Executives (Courtesy of Linnea Gandhi):

“Estimate the number of murders in:”

- the Netherlands
- Amsterdam



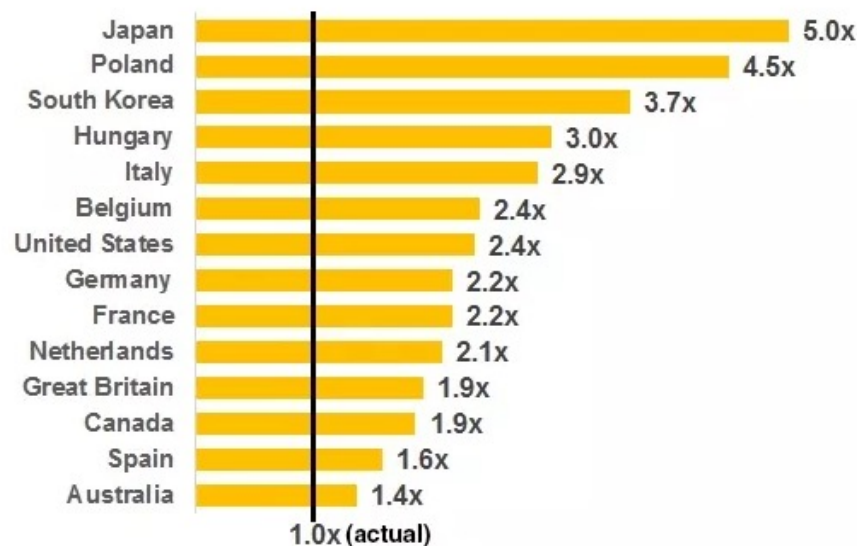
Availability I

- What percentage of the population do you think are:

Immigrants

"What percentage of the population do you think are immigrants?"

Avg. guess as a multiple of actual number

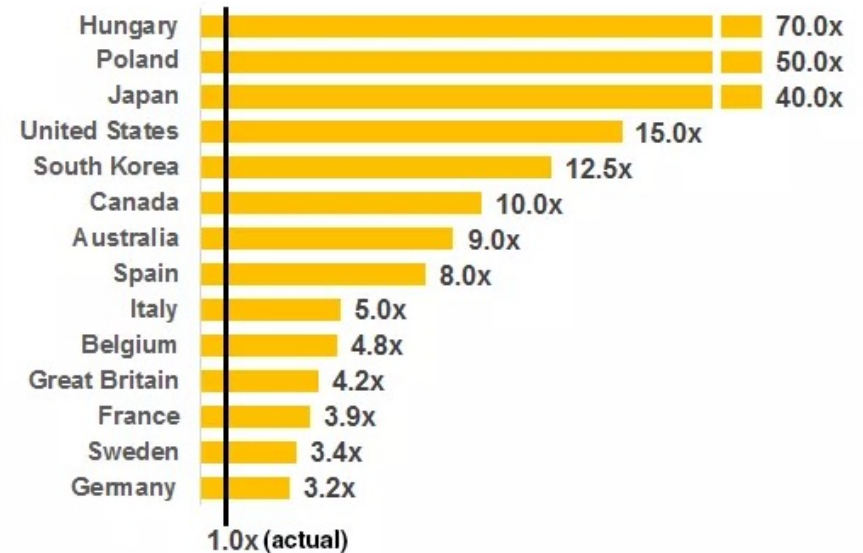


Source: IPSOS

Muslims

"Out of every 100 people how many do you think are Muslims?"

Avg. guess as a multiple of actual number



Source: IPSOS

Availability II

- Example 1 (False Consensus Effect)

- We use ourselves as a model in forming judgments about how others will behave.

% who will take 50-50 gamble?

Risk-takers (63%)

Risk-aversers (37%)

Salary in 10 years (Median: \$600K)

Below Median (\$425K)

Above Median (\$1000K)

Availability III

■ Example 2 (Reasoning by Recall)

<u>Cause of Death</u>	<u>Estimate (000s)</u>	<u>Actual (000s)</u>	<u>Ratio (Est/Act)</u>	<u>Google News search (6/13/16, 000s)</u>
Colon		52.9		
Fires	36	3	11.01	886
Homicide	54	20	2.70	970
Motor Vehicles	145	43	3.37	265
Pancreatic Cancer	50	30	1.65	80
Lung Cancer	69	161	0.43	385

Availability IV

■ Example 3 (Unpacking)

- What is the probability that it will rain in Chicago during the week of Monday, May 5-May 11, 2025 (inclusive)?

<u>Condition</u>	<u>Probability</u>		
Packed (rain)	56%		
Unpacked	90%	40%	(once)
		50%	(two or more)

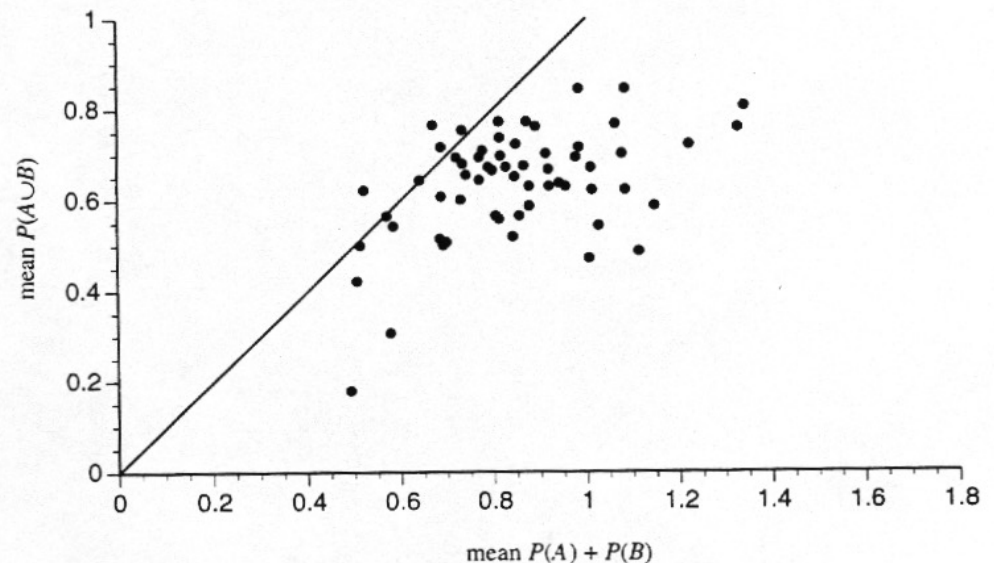
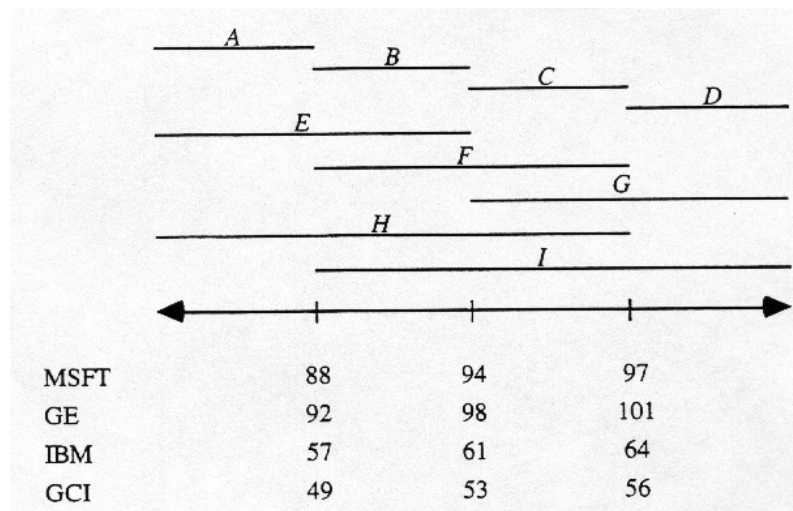
■ Example 4 (Unpacking)

- What is the probability that University of Chicago will be in the top 3 (1 / 2-3) in the 2026 *Business Week* rankings?

<u>Condition</u>	<u>Probability</u>		
Packed (top 3)	63%		
Unpacked	98%	35%	(first)
		58%	(2 or 3)

Unpacking Continued

- Normative Rule (Additivity of probabilities)
 - If A and B are mutually exclusive, then
$$\Pr(A \text{ or } B) = \Pr(A) + \Pr(B)$$
- Options Trader study (Fox, Rogers, and Tversky, 1995)
 - Subjects: options traders, Pacific Stock Exchange/ Chicago Board Options Exchange
 - Uncertain Quantities: Stock prices (Microsoft, General Electric, IBM, Gannett Company)



Availability

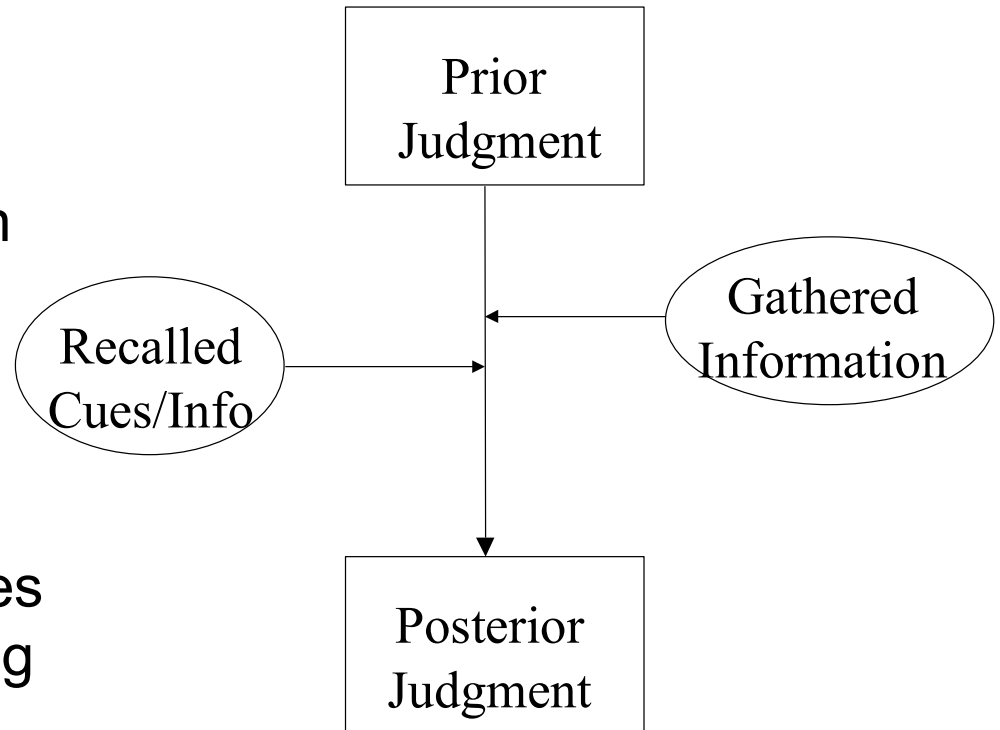
- In forming a judgment, we tend to
 - ❑ rely on readily available (recent, salient, vivid) information
 - ❑ but fail to discount the quality of the information, or consider other possible sources



Identify examples: Overweighted some recent (maybe atypical) outcome relative to another outcome

Updating Judgments

- Two processes
 - information collection/recall
 - weighting of information
- Normative Theory
 - Bayes Rule
- Descriptive Theory
 - Biases in both processes (weighting and gathering of information)

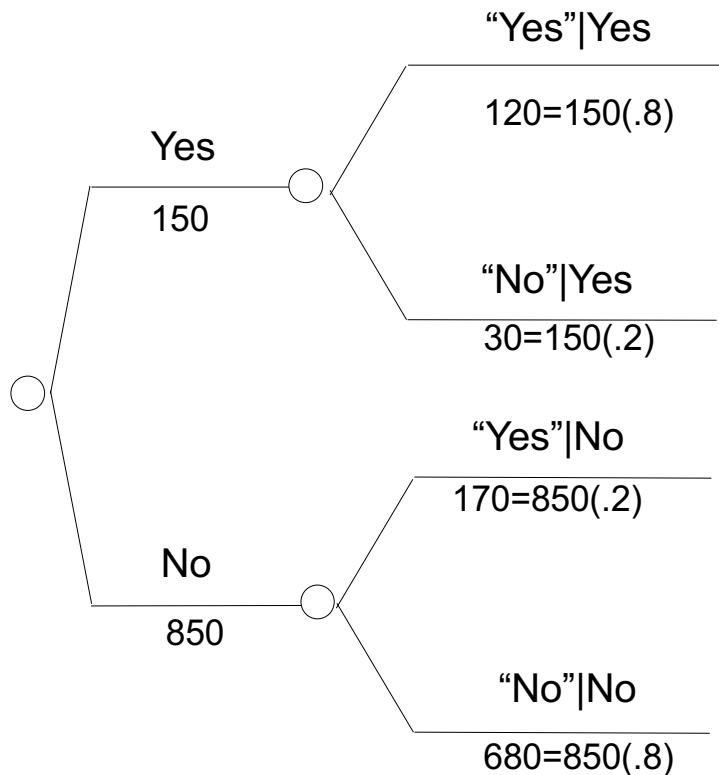


Updating Information

■ Problem

- ❑ A university uses an AI-detection tool to identify whether student essays are written by ChatGPT. When tested, the detector correctly identifies ChatGPT-written essays 80% of the time but incorrectly flags 20% of human-written essays as AI-generated.
- ❑ If the detector flags an essay as “written by ChatGPT,” what is the probability that it truly was written by ChatGPT?
- ❑ What additional information would you like to know?
 - The base rate! What percentage of students use ChatGPT!
 - Hypothetical Answer: 15%

ChatGPT Example: Normative Analysis



$$\begin{aligned}\Pr(\text{"Yes"}) &= \\ &\Pr(\text{"Yes"}|\text{No})\Pr(\text{No}) + \Pr(\text{"Yes"}|\text{Yes})\Pr(\text{Yes}) = \\ &.2(.85) + .8(.15) = \mathbf{.29}\end{aligned}$$

$$\begin{aligned}\Pr(\text{Yes}|\text{"Yes"}) &= \\ &\Pr(\text{"Yes"}|\text{Yes})\Pr(\text{Yes}) / \Pr(\text{"Yes"}) = .12 / .29 = \mathbf{.41}\end{aligned}$$

Normative Answer depends on

The base rate

$$\text{Ex 1: } \Pr(\text{Yes}) = .50 \Rightarrow \Pr(\text{Yes}|\text{"Yes"}) = .8$$

$$\text{Ex 2: } \Pr(\text{Yes}) = .85 \Rightarrow \Pr(\text{Yes}|\text{"Yes"}) = .96$$

The quality of the information

$$\begin{aligned}\text{Ex 3: } \Pr(\text{"Yes"}|\text{Yes}) &= \Pr(\text{"No"}|\text{No}) = .95 \Rightarrow \\ \Pr(\text{Yes}|\text{"Yes"}) &= .8\end{aligned}$$

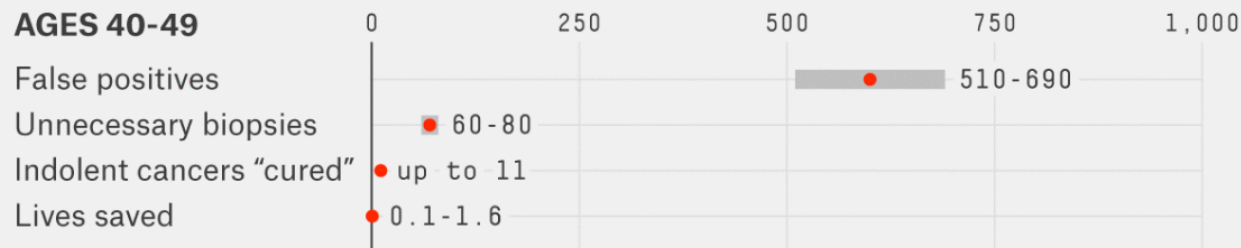
$$\begin{aligned}\text{Ex 4: } \Pr(\text{"Yes"}|\text{Yes}) &= \Pr(\text{"No"}|\text{No}) = .5 \Rightarrow \\ \Pr(\text{Yes}|\text{"Yes"}) &= .15\end{aligned}$$

A Controversial Topic: Mammograms?

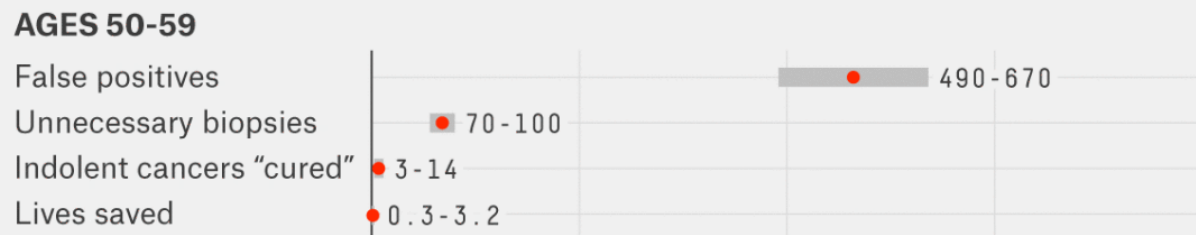
How Effective Are Mammograms?

Medical outcomes if 1,000 women receive annual mammogram exams, by age; estimate range comes from multiple studies

AGES 40-49



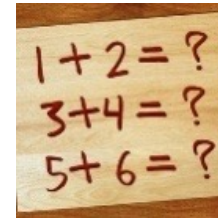
AGES 50-59



AGES 60-69



Optional Problem Set



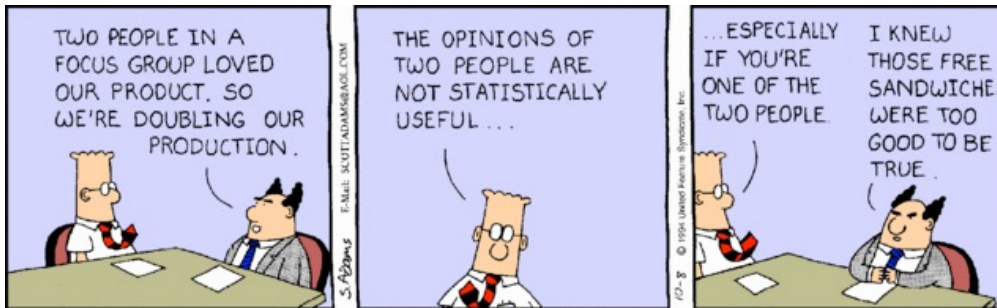
■ Problem

- "Abnormal" physical exam: What are the false positive ($\Pr(\text{No Cancer} \mid \text{Positive})$) and false negative ($\Pr(\text{Cancer} \mid \text{Negative})$) rates for mammograms if:
 - $\Pr(\text{Cancer}) = 8\%$
 - $\Pr(\text{Positive} \mid \text{Cancer}) = 92\%$
 - $\Pr(\text{Negative} \mid \text{No Cancer}) = 88\%$

- "Normal" physical exam:
 - $\Pr(\text{Cancer}) = 0.89\%$
 - $\Pr(\text{Positive} \mid \text{Cancer}) = 92\%$
 - $\Pr(\text{Negative} \mid \text{No Cancer}) = 88\%$

Biases in Updating Information

- Information is overweighted relative to base rate
 - Example: ChatGPT example
- Quality of the information (sample size; validity) is overweighted
 - Example 1

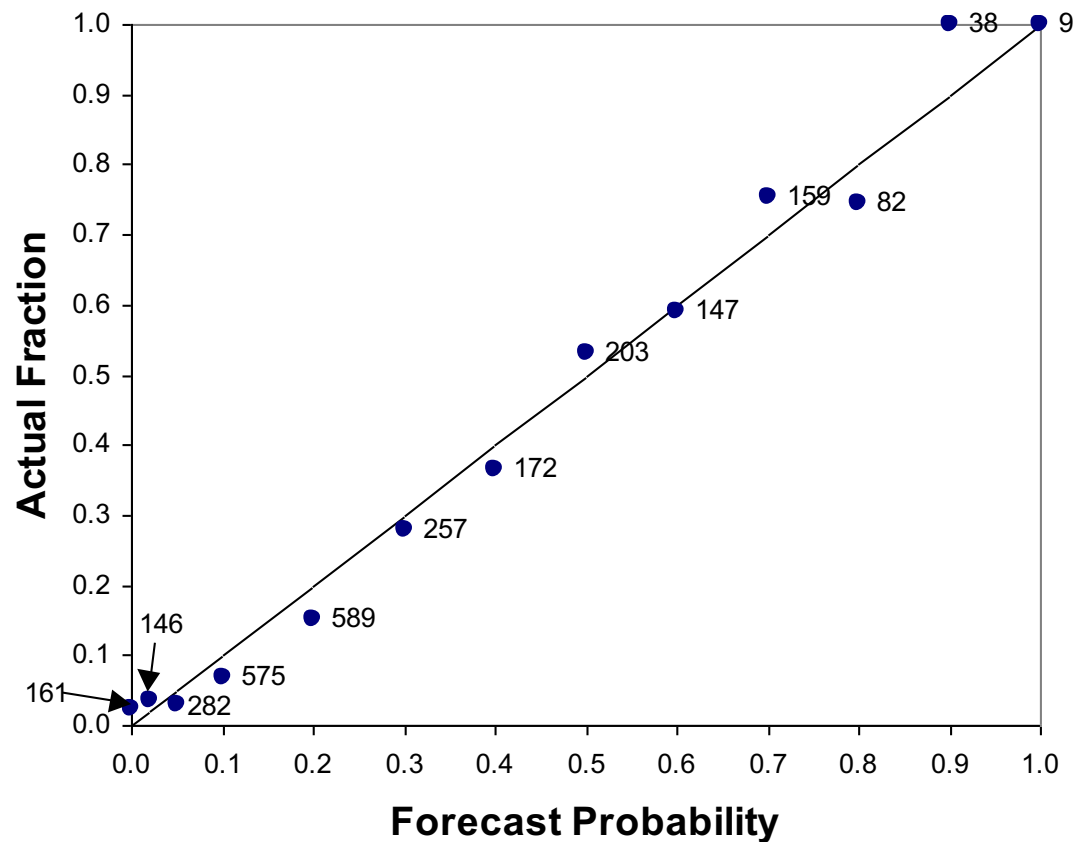


- Example 2: Buying a Lexus
- Example 3: Glowing recommendation
- Tendency to focus on information consistent with favored hypothesis, and ignore information consistent with other hypotheses
 - Example: *A Time to Kill*

Experts I: Weather Forecasters

2,820 Probability-of-Precipitation Forecasts by a National Weather Service Forecaster, Chicago, IL (July 1972-June 1976) (Murphy and Winkler, 1977) [Note: NWS forecasters have been giving probabilistic forecasts since 1965!]

Forecast Probability of Rain	Number of Forecasts	Number of Rainy Days	Fraction of Raining Days
0.00	161	4	0.025
0.02	146	5	0.034
0.05	282	8	0.028
0.10	575	39	0.068
0.20	589	89	0.151
0.30	257	72	0.280
0.40	172	63	0.366
0.50	203	108	0.532
0.60	147	87	0.592
0.70	159	120	0.755
0.80	82	61	0.744
0.90	38	38	1.000
1.00	9	9	1.000
	2,820	703	0.249



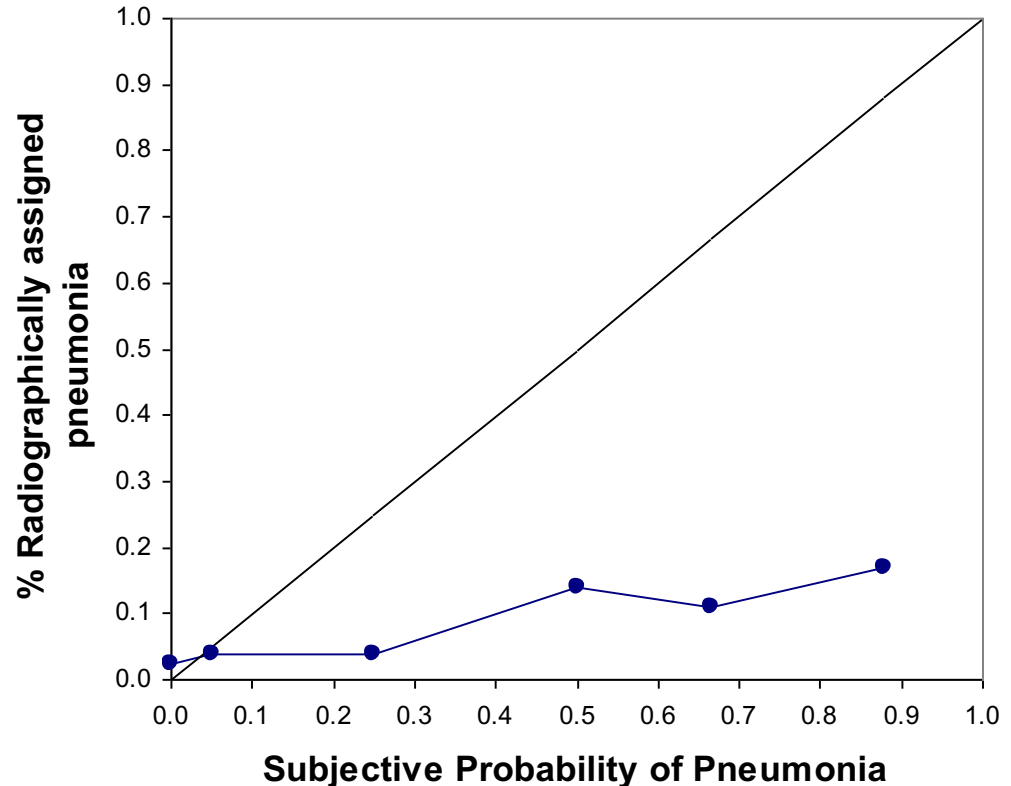
Experts II: Physicians

Physicians, after completing history and physical examination, estimated the probability that patients had pneumonia (Christensen-Szalanski & Bushyhead, 1981)

Why are weather forecasters so much better than physicians?



How timely is feedback on forecasts in your business?



Remedying Overconfidence

- Improving Secondary Knowledge: Process Suggestions
 - ❑ Start with extreme scenarios (to counteract anchoring). In a group, take the lowest of low and highest of high.
 - ❑ Be contrarian (“How could I be wrong?”) by using availability to your advantage:
 - “Tell me a story for why the DJIA is over 50,000 at the end of next year. Another... Another...”
 - “...under 35,000...”
 - ❑ In group, designate someone to be the contrarian
 - ❑ Scenario Analysis (see Russo and Schoemaker)

Remedying Overconfidence

- Seek out “reference class” or “base rate” information: Begin with the assumption that this situation is no different from the base rate
 - ❑ HK XRL forecast was initially 50% optimistic (4 years instead of 6 years)
-  Identify a forecasting situation that is part of a “reference class.”
What data could you collect to improve your forecast?
- Improving Secondary Knowledge: Organizational Suggestions
 - ❑ Simulate experience and monitor feedback (Shell Oil)
 - ❑ Use “prediction markets” to exploit “wisdom of crowds” (Google, Intel, etc)

Wu's Clues: Part 2



Clue 6: Many normative rules will never be learned through experience alone.

Clue 7: A real expert is someone who has outstanding primary knowledge, as well as excellent secondary knowledge.

Clue 8: Watch how you start, anchors and availability.

Clue 9: False confidence (i.e., overconfidence) is a decision maker's worst friend. The source of genuine confidence is a good decision making process.